

Row-Centring Fix and Pipeline Reorganisation

DPCN Assignment 1 — Opinion Network Formation

September 17, 2026

1 Scope of this note

This note documents a specific piece of work done on the repository: an audit of whether row-centring was implemented anywhere in the codebase, why it is required, and a subsequent reorganisation of the single-file pipeline into separate modules. It is not the assignment report; it is a record of one engineering change for the team.

2 What was found

Before this change, the only pipeline implemented in the repository (`src/ordinal_pipeline.py`) built a 60-node item–item network from *raw*, *uncentred* Spearman correlations computed on a matrix whose missing cells (structural blanks and “No Comments”) had been recovered by per-item regularised proportional-odds ordinal logistic regression. No row-centring step existed anywhere in the codebase or in the accompanying notebook.

This matters because the grand mean of the raw responses is $+1.14$ on a $[-2, +2]$ scale: respondents agree with nearly everything (acquiescence bias). A correlation computed on raw scores is therefore dominated by shared agreeableness rather than by which specific items a respondent leans toward or away from.

3 Why row-centring is required

For respondent i and item j , let $x_{i,j}$ denote the (imputed) encoded response on the $\{-2, -1, 0, 1, 2\}$ scale. Row-centring replaces $x_{i,j}$ with

$$x'_{i,j} = x_{i,j} - \bar{x}_i, \quad \bar{x}_i = \frac{1}{60} \sum_{k=1}^{60} x_{i,k}. \quad (1)$$

Subtracting each respondent’s own mean across all 60 items removes their personal baseline level of agreement before any correlation is computed. Spearman correlation does not do this automatically: it is invariant to a monotonic transform applied to a single column, not to an additive shift that differs row by row. Centring is applied strictly *after* missing-value imputation (imputation still predicts on the original -2 to $+2$ scale) and strictly *before* the correlation step, since it is a network-construction transform, not a data-cleaning one.

4 Empirical confirmation

Row-centring was added to the pipeline and run end-to-end on the real data (91 respondents after excluding entirely empty submissions, 60 items). Table 1 compares the item–item correlation matrix computed on the raw imputed matrix against the same matrix after row-centring, at the edge threshold used for the visualised network.

Matrix	Mean $ r $	Median $ r $	Edges at $ r \geq 0.30$
Raw (uncentred)	0.182	0.162	327
Row-centred	0.110	0.094	42

Table 1: Effect of row-centring on the item–item correlation matrix. Centring removes the acquiescence-bias component, collapsing the edge count at a fixed threshold by roughly 87%.

The raw network’s higher edge count is consistent with the near-complete, uninformative graph that raw-score similarity is expected to produce: most of those 327 edges reflect shared agreeableness rather than genuine item-to-item association. This is the same effect anticipated in the project’s frozen pipeline decisions, now verified against the actual dataset rather than assumed.

5 Scope note: what row-centring was *not* used to justify

No respondent-filtering threshold (e.g., dropping respondents with more than 12 missing items) was introduced as part of this change, in either direction. Two draft specifications considered such a filter; neither was applied. All 91 respondents retained after removing entirely empty submissions remain in the analysis, consistent with the repository’s existing behaviour prior to this change.

6 Repository reorganisation

The single 788-line `src/ordinal_pipeline.py` module was split into focused, importable modules:

- `src/config.py` — shared constants and `PipelineConfig`.
- `src/loader.py` — CSV parsing, missingness identification, encoding, and dataset export helpers.
- `src/imputation.py` — the ordinal logistic regression imputer, training, and masking-based validation.
- `src/similarity.py` — `row_centre()` (new) and the Spearman correlation computation, plus a raw-vs-centred comparison helper.
- `src/metrics.py` — graph construction and descriptive network statistics.
- `src/figures.py` — network visualisation.
- `src/reporting.py` — Markdown/JSON summary generation, extended to report the raw-vs-centred comparison.
- `src/pipeline.py` — orchestrates the above; imported by a new top-level `run_all.py`.

The notebook's import was updated to the new module paths, a `.gitignore` was added for `.venv/`, `__pycache__`, and notebook checkpoints, and `run_all.py` was verified to reproduce all outputs from a clean invocation.

7 Open items for the team

- The primary respondent–respondent network described in the project's frozen pipeline decisions (86 nodes, row-centred Pearson similarity, symmetrised k NN with $k = 4$, Louvain communities, degree-preserving null model, Deffuant–Weisbuch dynamics) is not yet implemented anywhere in the repository. This reorganisation only covers the item–item network and the imputation pipeline that already existed.
- An untracked draft specification file, `data/temp.md`, disagreed with the checked-in `CLAUDE.md` on which network is primary, which null model is primary, and the imputation statistic. That file was found missing from disk during this work; it was not removed by any command run as part of this change. Its content is recoverable if needed. This conflict should be resolved by the team before the primary network is built.